CLUSTER COMPUTING:

PROJECT REPORT

SENTIMENT ANALYSIS OF AMAZON REVIEWS USING PYSPARK

Submitted by Tenzin Choeyang Dhargyal Registration No – 12310784

Programme and Section – B.Tech. & K3P23GR

Course Code – INT 315

Under the Guidance of

Ma'am Zeenat Zahra & U.ID – 30447

Discipline of CSE/IT

Lovely School of Computer Science and Engineering

Lovely Professional University, Phagwara





CERTIFICATE

This is to certify that Tenzin Choeyang Dhargyal bearing Registration no. 12310784 has completed INT 315 project titled, “Sentiment Analysis of Amazon Reviews using PySpark” under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Ma’am Zeenat Zahra

Assistant Professor

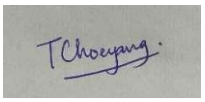
School of computer Science and Engineering

Lovely Professional University Phagwara, Punjab.

Date: 24-10-2026

DECLARATION

I, Tenzin Choeyang Dhargyal, student of B.Tech. under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.



Date: 24-10-2026
Registration No. 12310784 Signature

Acknowledgement

I would like to express my sincere gratitude to all those who have supported and guided me throughout the completion of this project.

First and foremost, I would like to thank Ma'am Zeenat Zahra for their valuable guidance, encouragement, and constant support during the course of this project. Their expertise and insightful suggestions greatly contributed to the success of this work.

Lastly, I extend my heartfelt thanks to my family and friends for their motivation, patience, and understanding throughout this journey.

Tenzin Choeyang Dhargyal
K3P23GR
12310784

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INTRODUCTION

Online product reviews are a rich source of customer opinion, but their volume makes manual reading impractical. Sentiment analysis addresses this by automatically classifying review text as positive, neutral or negative, so that sellers and buyers can quickly understand overall customer satisfaction.

This project uses Apache Spark through its Python API (PySpark) to build a sentiment classifier for Amazon product reviews. The work was carried out interactively in the PySpark shell started from the Windows Command Prompt. Review texts are converted into numerical features using a TF-IDF pipeline from Spark MLlib, and a Logistic Regression model is trained to predict the sentiment of unseen reviews. Sentiment labels are derived from the star rating given by the reviewer, and the model is evaluated using accuracy and weighted F1 score.

PROBLEM STATEMENT

The objective of this project is to design and implement a distributed machine-learning workflow that classifies Amazon reviews into positive, neutral and negative sentiment using only the review text.

The system aims to:

1. Load and clean a review dataset using Spark DataFrames.
2. Derive sentiment labels from the numeric rating.
3. Convert raw text into TF-IDF feature vectors using a reusable Spark ML pipeline.
4. Train and evaluate a Logistic Regression classifier.
5. Present the results in a clear, visual form.

OBJECTIVES

1. Create a SparkSession and load a CSV dataset containing multi-line text fields.
2. Inspect the schema and clean the data by casting types and removing null or empty records.
3. Create a sentiment column from the rating (overall).
4. Split the data into training and testing sets.
5. Build a pipeline consisting of Tokenizer, HashingTF, IDF and StringIndexer.
6. Train a Logistic Regression model on the training data.
7. Evaluate the model with accuracy and F1 score.
8. Visualise the sentiment distribution and model performance with Matplotlib.

TOOLS AND TECHNOLOGIES

Technology	Purpose
Python	Programming language
Apache Spark / PySpark	Distributed data processing and DataFrame API
Spark MLlib (pyspark.ml)	Feature extraction, pipeline, classification and evaluation
Matplotlib	Visualisation of results
Windows Command Prompt	Running the interactive PySpark shell

SOURCE OF DATASET

The project uses the file amazon_reviews.csv, a dataset of Amazon product reviews (source: https://huggingface.co/datasets/huggingfacelearners/amazon-reviews-sentiment-analysis). The raw file contains **4,915 records** and 12 columns.

Column	Description
_c0	Row index (the CSV header for this column is blank)
reviewerName	Name of the reviewer
overall	Star rating given by the reviewer (1–5)
reviewText	Full text of the review
reviewTime	Date of the review
day_diff	Number of days since the review was written
helpful_yes	Number of users who found the review helpful
helpful_no	Number of users who found the review not helpful
total_vote	Total helpfulness votes
score_pos_neg_diff	Difference between positive and negative helpfulness votes
score_average_rating	Average helpfulness score
wilson_lower_bound	Wilson lower-bound score of review helpfulness

For this project only reviewText and overall are used. The helpfulness-related columns are not needed for sentiment classification.

DATA LOADING & PREPROCESSING

1. Creating the Spark Session and Loading the Data

```
>>> from pyspark.sql import SparkSession
>>> spark=SparkSession.builder.appName("AmazonReviewAnalysis").getOrCreate()
26/10/21 14:19:43 WARN SparkSession: Using an existing Spark session; only runtime SQL configurations will take effect.
>>> csv_path="C:\\Users\\Tenzin Choeyang D\\OneDrive\\Desktop\\spark\\amazon_reviews.csv"
>>> df=spark.read.csv(csv_path,header=True,inferSchema=False,quote='"',escape='"',multiLine=True)
>>> df.printSchema()
root
 |-- _c0: string (nullable = true)
 |-- reviewerName: string (nullable = true)
 |-- overall: string (nullable = true)
 |-- reviewText: string (nullable = true)
 |-- reviewTime: string (nullable = true)
 |-- day_diff: string (nullable = true)
 |-- helpful_yes: string (nullable = true)
 |-- helpful_no: string (nullable = true)
 |-- total_vote: string (nullable = true)
 |-- score_pos_neg_diff: string (nullable = true)
 |-- score_average_rating: string (nullable = true)
 |-- wilson_lower_bound: string (nullable = true)

>>> df.count()
26/10/21 14:23:40 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
4915
```

The quote, escape and multiLine options let Spark read review texts that contain commas, quotation marks and line breaks without splitting a review across several rows. With inferSchema=False, every column is read as a string. df.count() returned **4,915** rows.

```
>>> df.show(5,truncate=80)
26/10/21 14:31:18 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
+---+------------+-------+--------------------------------------------------------------------------------+----------+--------+-----------+----------+----------+------------------+--------------------+--------------------+
|_c0|reviewerName|overall|                                                                      reviewText|reviewTime|day_diff|helpful_yes|helpful_no|total_vote|score_pos_neg_diff|score_average_rating|wilson_lower_bound|
+---+------------+-------+--------------------------------------------------------------------------------+----------+--------+-----------+----------+----------+------------------+--------------------+--------------------+
|  0|        NULL|    4.0|                                                                     No issues.|2014-07-23|     138|          0|         0|         0|                 0|                 0.0|                 0.0|
|  1|       0mie|    5.0|Purchased this for my device, it worked as advertised. You can never have too...|2013-10-25|     409|          0|         0|         0|                 0|                 0.0|                 0.0|
|  2|         1K3|    4.0|it works as expected. I should have sprung for the higher capacity.  I think ...|2012-12-23|     715|          0|         0|         0|                 0|                 0.0|                 0.0|
|  3|         1m2|    5.0|This think has worked out great.Had a diff. bran 64gb card and if went south ...|2013-11-21|     382|          0|         0|         0|                 0|                 0.0|                 0.0|
|  4|2&amp;1/2Men|    5.0|Bought it with Retail Packaging, arrived legit, in a orange envelope, english...|2013-07-13|     513|          0|         0|         0|                 0|                 0.0|                 0.0|
+---+------------+-------+--------------------------------------------------------------------------------+----------+--------+-----------+----------+----------+------------------+--------------------+--------------------+
only showing top 5 rows
```

First five rows of the raw data.

2. Cleaning the Data

```
>>> df=df.withColumn("overall",col("overall").cast("double"))
>>> from pyspark.sql.functions import col,when
>>> df=df.select("reviewText","overall")
>>> df=df.withColumn("overall",col("overall").cast("double"))
>>> df=df.filter(col("reviewText").isNotNull() & (col("reviewText")!="") & col("overall").isNotNull())
```

The cleaning steps were:

1. The overall rating was cast from string to double.
2. Only the two required columns, reviewText and overall, were kept.
3. Rows with a missing or empty review text or a missing rating were removed.

After cleaning, **4,914** rows remained, so one invalid record was removed.

SENTIMENT LABELLING

The dataset has no sentiment column, so labels were derived from the star rating: **Rating (overall) Sentiment**

1 or 2	negative
3	neutral
4 or 5	positive

```
>>> df = df.withColumn("sentiment", when(col("overall") <= 2, "negative").when(col("overall") == 3, "neutral").when(col("overall") >= 4, "positive"))
>>> df.show(5,truncate=80)
26/10/21 14:42:51 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
+--------------------------------------------------------------------------------+-------+---------+
|                                                                      reviewText|overall|sentiment|
+--------------------------------------------------------------------------------+-------+---------+
|                                                                      No issues.|    4.0| positive|
|Purchased this for my device, it worked as advertised. You can never have too...|    5.0| positive|
|it works as expected. I should have sprung for the higher capacity.  I think ...|    4.0| positive|
|This think has worked out great.Had a diff. bran 64gb card and if went south ...|    5.0| positive|
|Bought it with Retail Packaging, arrived legit, in a orange envelope, english...|    5.0| positive|
+--------------------------------------------------------------------------------+-------+---------+
only showing top 5 rows
```

The class distribution was obtained with df.groupBy("sentiment").count().show():

```
>>> df.count()
26/10/21 14:43:06 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
4914
>>> df.groupby("sentiment").count().show()
26/10/21 14:43:25 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
+---------+-----+
|sentiment|count|
+---------+-----+
| positive| 4448|
|  neutral|  142|
| negative|  324|
+---------+-----+
```

The dataset is strongly imbalanced: more than nine out of ten reviews are positive.

TRAIN - TEST SPLIT

The data was randomly divided into 80% training and 20% testing data, using a fixed seed so the split can be reproduced:

```
>>> train_df,test_df=df.randomSplit([0.8,0.2],seed=42)
>>> print("Training rows:",train_df.count())
26/10/21 14:45:34 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_a
verage_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
e_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
Training rows: 3976
```

Training Dataset

```
>>> print("Testing rows:",test_df.count())
26/10/21 14:45:52 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_a
verage_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
e_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
Testing rows: 938
```

Testing Dataset

FEATURE ENGINEERING PIPELINE

Machine-learning algorithms need numerical input, so the text is converted into TF-IDF vectors using a Spark ML Pipeline with four stages:

Stage	Role
Tokenizer	Splits each review into lower-case words
HashingTF	Maps words to a fixed-size vector of 10,000 term-frequency features
IDF	Down-weights words that appear in many reviews and highlights distinctive ones
StringIndexer	Converts the text sentiment into a numeric label (positive = 0.0)

```
>>> from pyspark.ml import Pipeline
>>> from pyspark.ml.feature import Tokenizer,HashingTF,IDF,StringIndexer
>>> tokenizer=Tokenizer(inputCol="reviewText",outputCol="words")
>>> hashingTF=HashingTF(inputCol="words",outputCol="rawfeatures",numFeatures=10000)
>>> idf=IDF(inputCol="rawfeatures",outputCol="features")
>>> indexer=StringIndexer(inputCol="sentiment",outputCol="labels")
>>> pipeline=Pipeline(stages=[tokenizer,hashingTF,idf,indexer])
```

The pipeline was fitted on the training data only, and the fitted model was then used to transform both sets, so that no information from the test data influenced the feature calculation:

```
>>> tfidf_model=pipeline.fit(train_df)
26/10/21 14:50:29 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, s
ore_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
26/10/21 14:50:34 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
_average_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, s
ore_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
```

```
>>> train_df=tfidf_model.transform(train_df)
>>> test_df=tfidf_model.transform(test_df)
>>> train_df.select("reviewText","sentiment","labels","features").show(5,truncate=60)
26/10/21 14:52:42 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_ratin
g, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_average_ra
ting, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
+------------------------------------------------------------+---------+------+------------------------------------------------------------+
|                                                  reviewText|sentiment|labels|                                                    features|
+------------------------------------------------------------+---------+------+------------------------------------------------------------+
|$20. less money than comparable Samsung. Very fast. Insta...| positive|   0.0|(10000,[371,488,679,1195,1756,2016,2911,2993,3228,3261,36...|
|**Update 6.24.14**My 128 GB card stopped allowing writes ...| positive|   0.0|(10000,[19,30,55,63,66,73,78,86,125,222,281,292,298,307,3...|
|  . . . . . . . . . . . . . . . no issues till date so happy| positive|   0.0|(10000,[1611,2313,3704,5347,6403,6712,6828],[2.6534934420...|
|... Got this to use in the Cobra dashcam. The cam came wi...| positive|   0.0|(10000,[222,488,750,815,1299,1504,2144,2288,2429,2605,265...|
|...and they're right!This microSDHC card is fast. I start...| positive|   0.0|(10000,[93,690,804,1152,1310,1420,1590,1756,1844,2094,214...|
+------------------------------------------------------------+---------+------+------------------------------------------------------------+
only showing top 5 rows
```

Transformed training data showing reviewText, sentiment, labels and the sparse features vector.

MODEL TRAINING

A multinomial Logistic Regression classifier was trained on the TF-IDF features:

```
>>> from pyspark.ml.classification import LogisticRegression
>>> lr=LogisticRegression(featuresCol="features",labelCol="labels",maxIter=50)
>>> lr_model=lr.fit(train_df)
26/10/21 14:54:52 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_a
verage_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
e_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
26/10/21 14:54:53 WARN InstanceBuilder: Failed to load implementation from:dev.ludovic.netlib.blas.JNIBLAS
26/10/21 14:54:53 WARN CSVHeaderChecker: CSV header does not conform to the schema.
 Header: , reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, score_a
verage_rating, wilson_lower_bound
 Schema: _c0, reviewerName, overall, reviewText, reviewTime, day_diff, helpful_yes, helpful_no, total_vote, score_pos_neg_diff, scor
e_average_rating, wilson_lower_bound
Expected: _c0 but found:
CSV file: file:///C:/Users/Tenzin%20Choeyang%20D/OneDrive/Desktop/spark/amazon_reviews.csv
```

Logistic Regression was chosen because it is fast, works well with high-dimensional sparse text features and is natively supported in Spark MLlib. Spark printed a JNIBLAS warning when training started, which only means that it used its pure-Java linear algebra library instead of a native one; it does not affect.

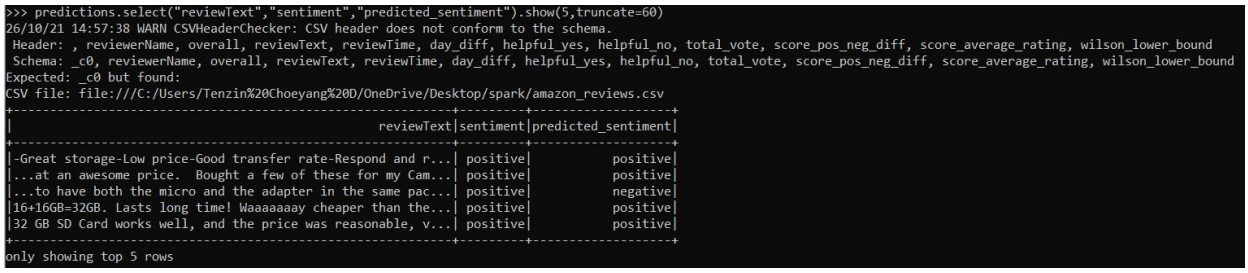
PREDICTION & EVALUATION

1. Predictions

The trained model was applied to the test set, and numeric predictions were converted back to text using IndexToString with the label mapping learned by the StringIndexer:

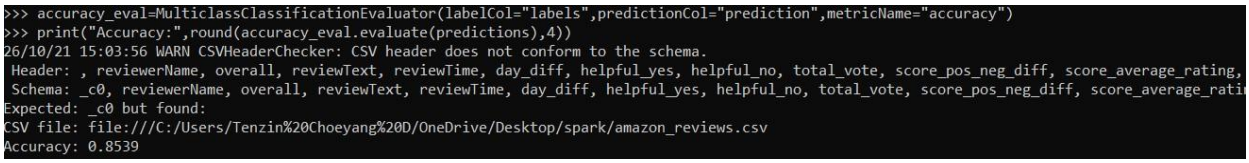
```
>>> from pyspark.ml.feature import IndexToString
>>> from pyspark.ml.evaluation import MulticlassClassificationEvaluator
>>> predictions=lr_model.transform(test_df)
>>> convertor=IndexToString(inputCol="prediction",outputCol="predicted_sentiment",labels=tfidf_model.stages[3].labels)
>>> predictions=convertor.transform(predictions)
```

In the five sample rows displayed, four reviews were classified correctly as positive and one positive review was misclassified as negative.

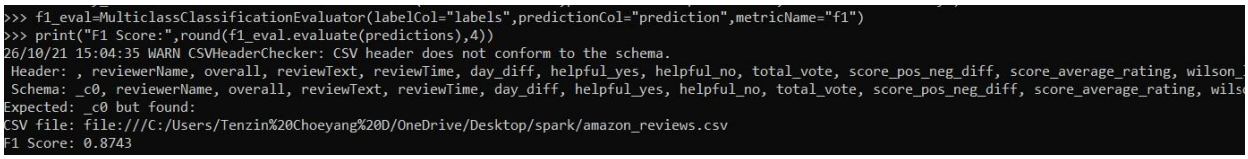


2. Evaluation Metrics

The model was evaluated with MulticlassClassificationEvaluator:



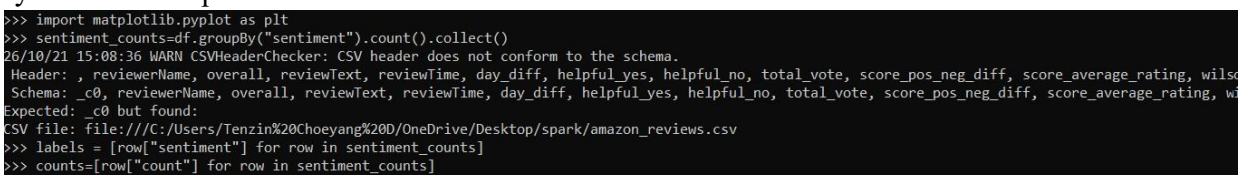
Accuracy: 0.85



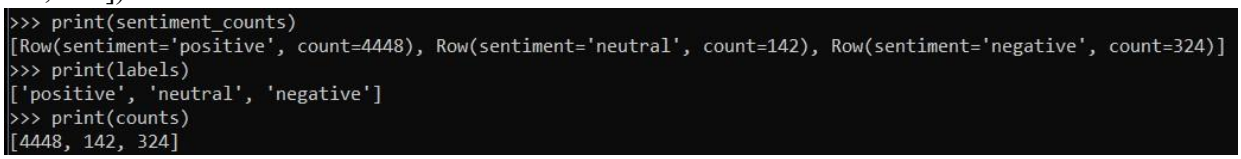
F1 Score:0.87

VISUALIZATION

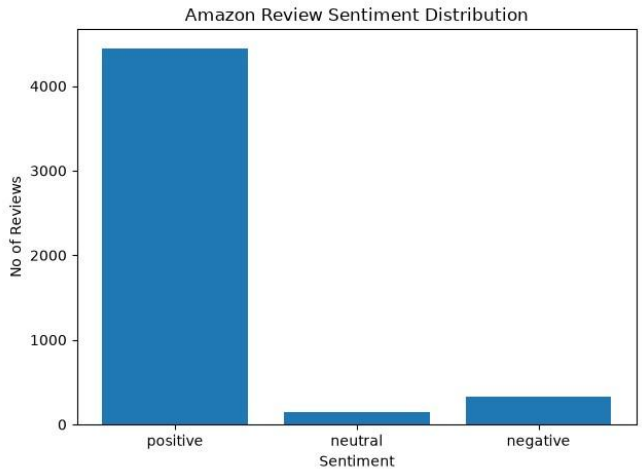
Matplotlib was used to visualise the results. The sentiment counts were collected from Spark into Python lists and plotted as a bar chart:



Code preparing the sentiment counts and printed lists (['positive', 'neutral', 'negative'] and [4448, 142, 324]).

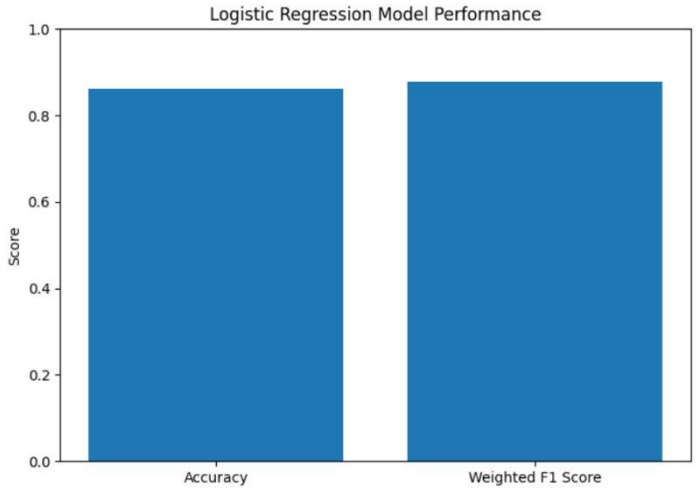


```
>>> plt.bar(labels,counts)
<BarContainer object of 3 artists>
>>> plt.title("Amazon Review Sentiment Distribution")
Text(0.5, 1.0, 'Amazon Review Sentiment Distribution')
>>> plt.xlabel("Sentiment")
Text(0.5, 0, 'Sentiment')
>>> plt.ylabel("No of Reviews")
Text(0, 0.5, 'No of Reviews')
>>> plt.tight_layout()
>>> plt.show()
```



Bar chart "Amazon Review Sentiment Distribution". The chart shows a very tall positive bar compared with the neutral and negative bars.

```
plt.figure(figsize=(7, 5))
plt.bar(["Accuracy", "Weighted F1 Score"], [accuracy_score, f1_score])
plt.title("Logistic Regression Model Performance")
plt.ylabel("Score")
plt.ylim(0, 1)
plt.tight_layout()
plt.show()
```



Bar chart compares the model's accuracy and weighted F1 score on a 0 to 1 scale.

RESULTS AND DISCUSSION

Metric	Value
Raw reviews	4,915
Reviews after cleaning	4,914
Training / testing rows	3,976 / 938
Positive / Negative / Neutral	4,448 (90.5%) / 324 (6.6%) / 142 (2.9%)
Accuracy	0.8539 (85.39%)
Weighted F1 score	0.8743 (87.43%)

- **Data:** 4,915 raw reviews were reduced to 4,914 after removing a missing or empty record.
- **Sentiment distribution:** 90.5% of reviews are positive, 6.6% negative and 2.9% neutral, so customers are overwhelmingly satisfied with the product.
- **Model performance:** The Logistic Regression model reached an accuracy of 85.39% and a weighted F1 score of 87.43% on 938 unseen reviews.
- **Interpreting accuracy:** Because 90.5% of the reviews are positive, a model that always predicted "positive" would be right about 90% of the time. The accuracy therefore has to be read together with the weighted F1 score, and per-class results would be needed to judge how well the model recognises the neutral and negative minorities.
- **Misclassification:** Some positive reviews were predicted as negative. Reviews often mix praise with complaints, and a rating-based label does not always match the tone of the text.

Limitations:

1. The labels come from star ratings, not from human reading of the text.
2. The classes are heavily imbalanced, and neutral reviews are rare (142 in total).
3. Only one model and one set of parameters was tried, and no cross-validation was used.
4. TF-IDF treats words independently and ignores word order and context.

CONCLUSION

The project demonstrates an end-to-end sentiment analysis workflow in PySpark: loading multi-line CSV data, cleaning it with DataFrame operations, deriving sentiment labels, building a TF-IDF feature pipeline, training a Logistic Regression model and evaluating and visualising the results. The classifier achieved 85.39% accuracy and an 87.43% weighted F1 score, showing that Spark MLlib can be used to build a text classifier through a short, reusable pipeline. The workflow can scale to much larger review collections because the same code runs on a Spark cluster.

FUTURE SCOPE

1. **Handle class imbalance:** Use class weights, over-sampling or under-sampling to improve neutral and negative predictions.
2. **Richer evaluation:** Report precision, recall and a confusion matrix for every class.
3. **Hyperparameter tuning:** Use CrossValidator with a parameter grid for regularisation and the number of features.
4. **Other models:** Compare with Naive Bayes, Random Forest and Linear SVM.
5. **Better text processing:** Remove stop words, use n-grams and apply word embeddings or transformer models.
6. **Save and deploy:** Save the pipeline and model and serve them through an API or Spark Streaming for live reviews.
7. **Use more data:** Train on a larger dataset to improve generalisation.

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